COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 5 : SLO – 2 | Signed magnitude subtraction
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Multiple Choice Questions (5)

• 1. b — Sign
• 2. b — -2 ((+3)-(+5): smaller minus larger → negative, magnitude 5-3=2)
• 3. c — (+A) - (-B) (becomes addition of magnitudes, which can overflow)
• 4. b — +2 ((-3)-(-5) = +2: magnitudes 5-3=2, sign of larger magnitude operand is +)
• 5. b — -7 to +7

Two-Mark Questions (2)

• 1. (+A)-(-B) and (-A)-(+B) turn into magnitude ADDITION (subtracting a negative). The result magnitude A+B can exceed the (N-1)-bit magnitude field (e.g. 5+3=8 > 7 in 4-bit), so the true result cannot be represented — overflow.
• 2. Same-sign subtraction steps: (i) compare magnitudes; (ii) subtract the smaller magnitude from the larger; (iii) keep the common sign if the first operand's magnitude is larger, otherwise take the opposite sign; (iv) if magnitudes are equal the result is +0.

Scenario-Based Long Answer (1)

• (-5) - (+3): different signs → this becomes magnitude addition: 5 + 3 = 8, and the sign of the result is the sign of the first operand (negative) → true result -8.
• Step-by-step: -5 = 1101, +3 = 0011. Magnitudes 101 + 011 = 1000 (8). Magnitude field is 3 bits (max 111 = 7). 8 > 7 → magnitude cannot fit.
• OVERFLOW occurs: -8 is outside the 4-bit signed magnitude range (-7 to +7).

Match the Following (5)

• A. MSB — 1. Sign bit
• B. (+A) - (-B) — 2. Addition; possible overflow
• C. (-A) - (+B) — 3. Addition; same sign result
• D. (-3) - (-5) — 4. +2 (result)
• E. 4-bit signed magnitude — 5. Range from -7 to +7

Fill in the Blanks (5)

• 1. MSB (most significant / sign) 2. subtract/compare 3. (N-1)-bit magnitude 4. +8 (overflow) 5. less efficient (harder/more complex)

Scenario-Based Small Activity (1)

• (+5) - (+3): both positive. Compare magnitudes: 101 > 011. Subtract: 101 - 011 = 010 (2). Sign stays positive.
• Final result: 0010 = +2.

True or False (5)

• 1. FALSE — MSB stores the sign, not magnitude. 2. TRUE
• 3. TRUE — (-5)-(-3) = -5+3 = -2. 4. FALSE — (+3)-(+5) = -2, not +2.
• 5. FALSE — Two's complement has the simpler hardware.